

Batch Attribute Inference for Hall–Petch Constant Recovery

Relaxed .xyz Grain-Scale Sampling, Regression-Derived Lattice Stress, and Generalized Material Attribute Extraction

VSEPR-SIM Development Day #62

From VSIM Batch Orchestration to Microstructure-Derived Strength Inference

VSEPR-SIM Development

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Abstract

This paper introduces a batch attribute inference layer for VSEPR-SIM and VSIM scripting, using Hall–Petch constant recovery as the first formal demonstration case. The central purpose is to show that once a set of relaxed `.xyz` samples has been produced across controlled microstructural variation, many material attributes become recoverable through direct measurement, transformation, and regression rather than through hardcoded material-property assignment.

The Hall–Petch relationship,

$$\sigma_y = \sigma_0 + k_y d^{-1/2},$$

is used as the primary example. Here, σ_y is yield strength, d is average grain size, σ_0 is the extrapolated lattice friction stress, and k_y is the Hall–Petch strengthening coefficient. A single material sample cannot determine σ_0 , because one measurement of σ_y at one grain size gives only one point on the strengthening curve. Instead, σ_0 must be recovered from a batch of samples with different grain sizes by plotting σ_y against $d^{-1/2}$ and extracting the y -intercept of the resulting linear fit.

This work defines the notation, data pathway, and conceptual architecture required for VSIM to perform that recovery from relaxed `.xyz` sample batches. The result is not merely a Hall–Petch module, but the beginning of a more general batch attribute inference architecture where final-state samples can be analyzed for grain size, defect density, porosity, packing state, structural class, transport behavior, and regression-derived material constants.

1 Introduction

VSEPR-SIM has increasingly moved from direct particle-state generation toward analysis-driven material interpretation. Earlier development stages established that `.xyz`, `.xyzFull`, and related state files should contain ground-truth structural and trajectory information, while derived material properties should be extracted afterward by analysis modules. This principle prevents the simulation state from becoming contaminated by inferred quantities.

The present work extends that doctrine into a batch-level material inference system. Instead of treating one relaxed sample as a complete material description, the system treats a controlled collection of relaxed samples as a measurable ensemble. Each sample contributes one or more attributes. These attributes can then be transformed, fitted, compared, and used to infer higher-level material constants.

The Hall–Petch relation provides an ideal first target because it is simple, physically meaningful, and impossible to recover from a single sample. This makes it a useful test case for VSIM’s transition from single-sample analysis toward batch-based material characterization.

Core claim:

A relaxed `.xyz` batch can become a material inference dataset.

For Hall–Petch analysis, the path is:

$$\text{relaxed .xyz samples} \rightarrow (d_i, \sigma_{y,i}) \rightarrow d_i^{-1/2} \rightarrow \sigma_y = \sigma_0 + k_y d^{-1/2} \rightarrow (\sigma_0, k_y). \quad (1)$$

The `.xyz` files do not need to store the final constants. They only need to provide sample structure. The constants are recovered by the analysis layer.

2 Background

2.1 Hall–Petch Strengthening

The Hall–Petch relationship describes the increase in yield strength as grain size decreases:

$$\sigma_y = \sigma_0 + k_y d^{-1/2} \quad (2)$$

In this expression:

- σ_y is the yield strength of the material,
- σ_0 is the lattice friction stress, sometimes interpreted as the extrapolated yield strength for very large grain size,
- k_y is the Hall–Petch slope or grain-boundary strengthening coefficient,
- d is the average grain diameter.

The term $d^{-1/2}$ increases as grain size decreases. Therefore, smaller grains generally correspond to higher yield strength over the normal Hall–Petch regime.

The important mathematical point is that σ_0 is not directly obtained from one test. One sample gives:

$$\sigma_{y,1} = \sigma_0 + k_y d_1^{-1/2}. \quad (3)$$

This is one equation with two unknowns: (σ_0, k_y) . Therefore, multiple samples with different grain sizes are required.

For a batch of samples

$$(d_1, \sigma_{y,1}), (d_2, \sigma_{y,2}), \dots, (d_n, \sigma_{y,n}), \quad (4)$$

the transform $x_i = d_i^{-1/2}$ creates a linear regression problem:

$$\sigma_{y,i} = \sigma_0 + k_y x_i. \quad (5)$$

The intercept of the fitted line is σ_0 , and the slope is k_y .

3 Purpose of the VSIM Hall–Petch Inference Layer

The purpose of this development layer is to introduce a clean VSIM notation for batch attribute recovery from relaxed `.xyz` samples.

Specific Hall–Petch objective: recover σ_0 and k_y from a batch of relaxed grain-structured samples.

Broader architectural objective: define a reusable attribute-inference pipeline for future material constants.

This matters because many material attributes are not meaningful as direct fields inside a single state file. They are instead inferred from:

- structural measurements,
- metadata,
- controlled sample variation,
- fitted relationships,
- statistical or regression-based analysis.

The Hall–Petch module therefore becomes the first clean demonstration of a larger VSIM idea:

$$\text{sample batch} \rightarrow \text{attributes} \rightarrow \text{transforms} \rightarrow \text{fit} \rightarrow \text{material constants.} \quad (6)$$

This is the same conceptual direction needed for Arrhenius fits, diffusion activation energy, porosity-strength trends, defect-density relationships, packing-property inference, and other higher-order material models.

4 File Doctrine

4.1 .xyz as Final Relaxed Structure

For this inference layer, `.xyzFull` is not required. The system only needs final relaxed structures. The `.xyz` file is sufficient when the objective is to measure final geometry or connect a final sample to external metadata.

A relaxed `.xyz` sample should contain ground-truth structural data such as:

- particle identity,
- position,
- sample geometry,
- final relaxed configuration.

It should *not* directly encode:

- σ_0 , k_y , R^2 ,
- Hall–Petch classification,
- macro material labels.

Those quantities belong to the analysis layer.

4.2 Metadata Sidecars

Yield strength may come from an external source, a prior simulation, a physical test, or a calibrated proxy model. Since `.xyz` should remain structurally clean, sample-level non-structural attributes may be supplied through metadata sidecars.

Example file pairing:

```
1 cu_grain_05um.xyz
2 cu_grain_05um.vmeta.json
```

Example metadata:

```
1 {
2   "sample_id": "cu_grain_05um",
3   "material": "Cu",
4   "relaxed": true,
5   "grain_size_um": 5.0,
6   "sigma_y_MPa": 185.2,
7   "temperature_K": 300,
8   "strain_rate_s^-1": 0.001,
9   "source": "simulation"
10 }
```

The .xyz file carries the structure. The sidecar carries sample-level metadata. The analysis layer combines them. This keeps the state doctrine clean while still allowing practical material inference. Shocking that the solution is “do not stuff everything into one file,” but civilization has fallen for less.

5 VSIM Notation

5.1 Minimal Hall–Petch Study Notation

The simplest Hall–Petch VSIM notation is:

```
1 [study]
2 type = "hall_petch"
3 name = "cu_grain_strength_sweep"
4
5 [input]
6 format = "xyz"
7 files = [
8     "samples/cu_grain_05um.xyz",
9     "samples/cu_grain_10um.xyz",
10    "samples/cu_grain_25um.xyz",
11    "samples/cu_grain_50um.xyz"
12 ]
13
14 [measure.grain]
15 method = "relaxed_microstructure"
16 grain_size = "mean_equivalent_diameter"
17 units = "um"
18
19 [measure.yield]
20 source = "metadata"
21 field = "sigma_y"
22 units = "MPa"
23
24 [fit.hall_petch]
25 x = "grain_size^(-1/2)"
26 y = "sigma_y"
27 model = "linear"
28 recover = ["sigma_0", "k_y", "r2", "residuals"]
29
30 [export]
31 table = true
32 plot = "sigma_y_vs_d_inv_sqrt"
33 report = true
```

This notation is readable but somewhat specific to Hall–Petch. It works well as a first demonstration but should not become the final generalized syntax.

5.2 General Batch Attribute Inference Notation

The preferred canonical notation is:

```
1 [study]
2 type = "batch_attribute_fit"
```

```

3 name = "hall_petch_recovery"
4
5 [samples]
6 format = "xyz"
7 path = "samples/material_A/grain_sweep/*.xyz"
8 sidecar = "vmeta.json"
9
10 [attributes]
11 required = ["grain_size", "yield_strength"]
12
13 [attribute.grain_size]
14 from = "geometry"
15 method = "mean_equivalent_grain_diameter"
16 symbol = "d"
17 units = "um"
18 kind = "derived"
19
20 [attribute.yield_strength]
21 from = "metadata"
22 field = "sigma_y"
23 symbol = "sigma_y"
24 units = "MPa"
25 kind = "measured"
26
27 [transform.d_inv_sqrt]
28 from = "grain_size"
29 expression = "d^(-1/2)"
30 symbol = "d_inv_sqrt"
31 units = "um^(-1/2)"
32
33 [fit.hall_petch]
34 model = "sigma_y = sigma_0 + k_y * d_inv_sqrt"
35 x = "d_inv_sqrt"
36 y = "yield_strength"
37 outputs = ["sigma_0", "k_y", "r2", "residuals"]
38
39 [export]
40 table = "hall_petch_table.csv"
41 report = "hall_petch_report.md"
42 plot = "hall_petch_fit.png"

```

This is the preferred long-term form because Hall-Petch becomes one instance of a broader batch attribute model.

6 Attribute System

The key abstraction is the **attribute**.

A VSIM attribute is any measurable, derived, proxy, or fitted quantity attached to a sample or batch.

The general form is:

```

1 [attribute.<name>]
2 from   = "<source>"
3 method = "<method>"
4 symbol = "<math_symbol>"
5 units  = "<units>"
6 kind   = "<measured | derived | proxy | fitted>"

```

6.1 Example: Grain Size

```

1 [attribute.grain_size]
2 from   = "geometry"
3 method = "mean_equivalent_grain_diameter"
4 symbol = "d"
5 units  = "um"
6 kind   = "derived"

```

This means grain size is not read as a hardcoded material property. It is derived from sample geometry.

6.2 Example: Yield Strength

```

1 [attribute.yield_strength]
2 from   = "metadata"
3 field  = "sigma_y"
4 symbol = "sigma_y"
5 units  = "MPa"
6 kind   = "measured"

```

This means the yield strength is supplied from metadata, physical testing, previous simulation, or external analysis.

6.3 Example: Proxy Yield Strength

If yield strength is estimated from a model rather than measured, it should be marked explicitly:

```

1 [attribute.yield_strength]
2 from   = "proxy_model"
3 method = "grain_boundary_density_strength_proxy"
4 symbol = "sigma_y"
5 units  = "MPa"
6 kind   = "proxy"

```

This distinction is important. A proxy value is not the same as measured yield strength. VSIM should preserve that distinction in reports and exported data.

7 Hall–Petch Pipeline

The Hall–Petch inference pipeline follows this sequence:

1. Load relaxed XYZ batch

2. Load sidecar metadata
3. Derive grain-size attributes
4. Build sample attribute table
5. Transform d into $d^{-1/2}$
6. Fit σ_y against $d^{-1/2}$
7. Recover σ_0 and k_y
8. Export table, plot, and report

Mathematically, for each sample i :

$$d_i = \text{measured grain size}, \quad (7)$$

$$\sigma_{y,i} = \text{measured or supplied yield strength}, \quad (8)$$

$$x_i = d_i^{-1/2}. \quad (9)$$

The fitted relation is:

$$\sigma_{y,i} = \sigma_0 + k_y x_i. \quad (10)$$

The recovered constants are:

$$\sigma_0 = \text{intercept}, \quad (11)$$

$$k_y = \text{slope}. \quad (12)$$

The residual for each sample is:

$$r_i = \sigma_{y,i} - (\sigma_0 + k_y x_i). \quad (13)$$

The coefficient of determination is:

$$R^2 = 1 - \frac{\sum_i r_i^2}{\sum_i (\sigma_{y,i} - \bar{\sigma}_y)^2}. \quad (14)$$

8 Example Batch Table

A representative Hall–Petch input table may appear as:

Sample ID	Grain Size d (μm)	$d^{-1/2}$ ($\mu\text{m}^{-1/2}$)	Yield Strength σ_y (MPa)
cu_grain_05um	5.0	0.447214	185.2
cu_grain_10um	10.0	0.316228	161.7
cu_grain_25um	25.0	0.200000	139.4
cu_grain_50um	50.0	0.141421	128.1

The fitted result may be reported as:

$$\sigma_y = 104.8 + 181.2 d^{-1/2}. \quad (15)$$

Therefore:

$$\sigma_0 = 104.8 \text{ MPa}, \quad (16)$$

$$k_y = 181.2 \text{ MPa} \cdot \mu\text{m}^{1/2}. \quad (17)$$

The report should also provide:

- R^2 ,
- residuals,
- sample count,
- grain-size range,
- validity warning.

9 Implementation Architecture

9.1 Proposed Header

The first implementation target should be:

`src/analysis/batch_attribute_inference.hpp`

Hall-Petch-specific helpers may be placed in:

`src/analysis/hall_petch_inference.hpp`

However, the generic batch layer should be primary. The Hall-Petch system should not become a one-off isolated tool.

9.2 Core Data Structures

```

1  struct AttributeValue {
2      std::string name;
3      std::string symbol;
4      std::string units;
5      std::string kind;
6      double      value;
7  };
8
9  struct SampleAttributeRecord {
10     std::string      sample_id;
11     std::string      xyz_path;
12     std::vector<AttributeValue> attributes;
13 };
14
15 struct TransformSpec {
16     std::string name;
17     std::string input_attribute;
18     std::string expression;
19     std::string symbol;
20     std::string units;
21 };
22
23 struct LinearFitResult {
24     double      intercept;
25     double      slope;
26     double      r_squared;
27     std::vector<double> residuals;
28 };
29
30 struct HallPetchResult {

```

```

31     double          sigma_0;
32     double          k_y;
33     double          r_squared;
34     std::vector<double> residuals;
35 };

```

9.3 Core Functions

```

1  std::vector<SampleAttributeRecord>
2  load_sample_attribute_batch(const std::string& path);
3
4  double
5  derive_mean_equivalent_grain_diameter(const XYZStructure& structure);
6
7  double
8  apply_attribute_transform(double value, const TransformSpec& transform);
9
10 LinearFitResult
11 fit_linear_model(const std::vector<double>& x,
12                 const std::vector<double>& y);
13
14 HallPetchResult
15 fit_hall_petch(const std::vector<SampleAttributeRecord>& samples);

```

The Hall–Petch fit is mathematically simple, but it should be surrounded by strong validation. A useful module with weak validation is just a bug generator with a nicer hat.

10 Validation Requirements

The Hall–Petch inference layer should reject or warn under several conditions.

10.1 Minimum Sample Count

At least three samples should be preferred. Two samples can define a line, but cannot meaningfully establish fit quality.

Recommended rule:

```

1  minimum_required_samples = 3
2  preferred_samples >= 4

```

10.2 Grain-Size Variation

The batch must contain more than one grain size. If all grain sizes are equal or nearly equal, the regression is invalid.

Required check:

$$\max(d) - \min(d) > \epsilon_d. \quad (18)$$

10.3 Positive Grain Size

All grain sizes must satisfy:

$$d_i > 0. \quad (19)$$

Otherwise $d^{-1/2}$ is invalid.

10.4 Yield Strength Availability

Each sample must provide a yield-strength value through either:

- metadata,
- external table,
- simulation-derived value,
- explicitly marked proxy model.

If the value is a proxy, the exported report must say so.

10.5 Hall–Petch Validity Range

The report should state that the fitted relation is only supported over the tested grain-size range.

For extremely small nanocrystalline grain sizes, ordinary Hall–Petch behavior may break down. VSIM should therefore include a warning when grain sizes fall below a configured threshold.

Example warning:

Hall–Petch linearity is only validated over the supplied grain-size range. Nanocrystalline regimes may show deviation or inverse Hall–Petch behavior.

11 Export Format

11.1 CSV Table

The CSV output should include:

```
1 sample_id,xyz_path,d_um,d_inv_sqrt_um^-1/2,sigma_y_MPa,predicted_sigma_y_MPa,
   residual_MPa
```

11.2 JSON Report

The JSON output should include:

```
1 {
2   "study": "hall_petch_recovery",
3   "model": "sigma_y = sigma_0 + k_y * d_inv_sqrt",
4   "sample_count": 4,
5   "grain_size_units": "um",
6   "yield_strength_units": "MPa",
7   "fit": {
8     "sigma_0": 104.8,
9     "k_y": 181.2,
10    "r_squared": 0.991
11  },
12  "validity": {
```

```

13     "grain_size_min_um": 5.0,
14     "grain_size_max_um": 50.0,
15     "warning": "Fit should not be extrapolated beyond tested grain-size range."
16 }
17 }

```

11.3 Markdown Report

The Markdown report should include:

```

1 # Hall-Petch Inference Report
2
3 Material: Cu
4 Study:    hall_petch_recovery
5 Samples:  4
6
7 Model:
8   sigma_y = sigma_0 + k_y d^(-1/2)
9
10 Recovered Constants:
11   sigma_0 = 104.8 MPa
12   k_y     = 181.2 MPa.um^(1/2)
13   R^2     = 0.991
14
15 Interpretation:
16   The fitted relation indicates positive grain-boundary strengthening.
17   Yield strength increases as grain size decreases across the tested range.
18
19 Validity:
20   This result is valid only over the supplied grain-size range.

```

12 Relationship to Existing VSEPR-SIM Architecture

This layer is consistent with previous VSEPR-SIM development principles.

The existing system already supports or has planned support for:

- diffusion inference,
- packing inference,
- porosity estimation,
- transport anisotropy,
- macro property reporting,
- batch orchestration.

Hall-Petch inference extends this family into mechanical property recovery.

The important architectural shift is that *the batch becomes the object of analysis*. A single sample may provide grain size and yield strength, but the material constants arise only from the batch.

This leads to a more general VSEPR-SIM rule:

Some material properties are not sample attributes. They are ensemble-fit attributes.

Examples include:

$$\sigma_0, k_y, Q, D_0, E_{\text{activation}}, \text{scaling exponents.} \quad (20)$$

These should be stored as analysis outputs, not sample-state fields.

13 Generalization Beyond Hall–Petch

The same notation can support other models.

13.1 Arrhenius Diffusion Fit

$$D = D_0 e^{-Q/RT} \quad (21)$$

Linearized form:

$$\ln D = \ln D_0 - \frac{Q}{R} \frac{1}{T}. \quad (22)$$

VSIM form:

```
1 [fit.arrhenius_diffusion]
2 model = "ln_D = ln_D0 - Q_over_R * inv_T"
3 x      = "inv_T"
4 y      = "ln_D"
5 outputs = ["D0", "Q", "r2", "residuals"]
```

13.2 Porosity–Strength Fit

$$\sigma = \sigma_{\text{dense}} (1 - \phi)^n. \quad (23)$$

VSIM form:

```
1 [fit.porosity_strength]
2 model = "sigma = sigma_dense * (1 - porosity)^n"
3 x      = "porosity"
4 y      = "strength"
5 outputs = ["sigma_dense", "n", "r2"]
```

13.3 Defect-Density Strengthening

$$\sigma_y = \sigma_0 + A \sqrt{\rho_d}. \quad (24)$$

VSIM form:

```
1 [fit.defect_strength]
2 model = "sigma_y = sigma_0 + A * sqrt(defect_density)"
3 x      = "sqrt_defect_density"
4 y      = "yield_strength"
5 outputs = ["sigma_0", "A", "r2"]
```

These are not separate parser worlds. They are different fit blocks using the same batch attribute machinery.

14 Proposed Work Order

WO-VSEPR-SIM-62D

Batch Attribute Inference and Hall–Petch Constant Recovery

Objective

Implement a VSIM-compatible batch attribute inference layer that can recover Hall–Petch constants from relaxed .xyz grain-size sample batches.

Scope

This work order includes:

- relaxed .xyz batch loading,
- optional metadata sidecar loading,
- grain-size attribute extraction,
- yield-strength attribute ingestion,
- transform support for $d^{-1/2}$,
- linear regression fitting,
- recovery of σ_0 , k_y , R^2 , and residuals,
- CSV, JSON, Markdown, and plot export support.

Non-Scope

This work order does not require:

- .xyzFull trajectory parsing,
- dynamic deformation simulation,
- direct dislocation dynamics,
- finite-element coupling,
- automatic true yield prediction from geometry alone.

Those can be added later. Trying to do all of them now would be a lovely way to drown a clean idea in feature sewage.

Success Criteria

The work order is complete when VSIM can run:

```
1 vsepr run hall_petch_recovery.vsim
```

and produce:

```
1 hall_petch_table.csv
2 hall_petch_fit.json
3 hall_petch_report.md
4 hall_petch_fit.png
```

with recovered values for σ_0 , k_y , R^2 , and residuals.

15 Conclusion

This paper defines a clean VSIM notation and analysis architecture for recovering Hall–Petch constants from relaxed `.xyz` sample batches. The essential idea is that a single sample can provide measured or derived attributes, but certain material constants only emerge from controlled variation across a batch.

For Hall–Petch analysis, the controlled variable is grain size. Each relaxed sample contributes $(d_i, \sigma_{y,i})$. The batch transforms grain size into $d_i^{-1/2}$, then fits:

$$\sigma_y = \sigma_0 + k_y d^{-1/2}. \quad (25)$$

The recovered intercept gives σ_0 , and the recovered slope gives k_y .

The architectural importance is larger than the Hall–Petch example itself. This work establishes a general VSIM batch attribute inference pattern:

$$\text{samples} \rightarrow \text{attributes} \rightarrow \text{transforms} \rightarrow \text{fit} \rightarrow \text{constants}. \quad (26)$$

This allows VSEPR-SIM to move toward a system where many material attributes become recoverable from structured batches of final relaxed states rather than manually assigned properties. Hall–Petch constant recovery is therefore both a useful mechanical-materials module and a foundation for broader sample-to-property inference.

One sample is an anecdote. A batch, fitted, is a measurement.